1. Weighted Median Consider n numbers $p_1, \dots, p_n$ with associated positive weights $w_1, \dots, w_n$ which satisfy that $\sum_{i=1}^{n} w_i = 1$. We say that p_k is a *weighted median* of the given numbers if $\sum_{p_i < p_k} w_i < \frac{1}{2}$ and $\sum_{p_i > p_k} w_i \leq \frac{1}{2}$. Describe an algorithm which finds a weighted median of given numbers in time $O(n)$. The input numbers are given in an arraywhich is not sorted.

2. Post Office Placement We are given n points $P = \{p_1, \dots, p_n\}$ which have associated positive weights $w_1, \dots, w_n$ satisfying $\sum_{i=1}^{n} w_i = 1$ and a distance function $d(p_i, p_j) > 0$ which gives the distance between any two points p_i and p_j. We want to place the post office at a point p^* which minimizes the value of $\sum_{i=1}^{n} w_i \, d(p^*, p_i)$.

(a) Argue that, if $P \in \mathbb{R}^n$ and $d(p_i, p_j) = |p_i - p_j|$, then the optimal choice for p^* is the weighted median of the points.

(b) What is an optimal choice of p^* if $P \in \mathbb{R}^{n \times 2}$ and $d(p_i, p_j) = |p_{i1} - y_1| + |p_{i2} - p_{j2}|$.

3. Nadiria In a previous life, you worked as a cashier in the lost Antarctican colony of Nadiria, spending the better part of your day giving change to your customers. Because paper is a very rare and valuable resource in Antarctica, cashiers were required by law to use the fewest bills possible whenever they gave change. Thanks to the numerological predilections of one of its founders, the currency of Nadiria was available in the following denominations: \$1, \$4, \$7, \$13, \$28, \$52, \$91, and \$365.

(a) Describe and analyze a recursive algorithm that computes, given an integer k, the minimum number of bills needed to make k Dream-Dollars. (Don't worry about making your algorithm fast; just make sure it's correct.)

(b) Describe a dynamic programming algorithm that computes, given an integer k, the minimum number of bills needed to make k Dream-Dollars. (This one needs to be fast.)

4. Library Say that we have a sequence of n books in alphabetical order—each with a width w_i and a height h_i—and a single library cabinet with several shelves of fixed width W but mutable height. We want to arrange the books so as to minimize the total height of the cabinet while preserving the alphabetical ordering.

5. Broadcasting in a Tree Suppose we need to broadcast a message to all the nodes in a rooted binary tree. Initially, only the root node knows the message. In a single round, any node that knows the message can forward it to at most one of its children. Design an algorithm to compute the minimum number of rounds required to broadcast the message to all nodes.